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MatterMind: A User-Friendly Material Design Platform Assisted by Large Language Model

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ABSTRACT

Machine learning and AI assistants are reshaping materials research; however, their day-to-day application in experimental and computational workflows is still inconsistent, limited by insufficient software documentation and software engineering practices, fragmented software engineering ecosystem and brittle integration between computational tools. Especially in the field of materials informatics, the lack of user-friendly tool interfaces has restricted the popularization of data-driven methods in daily scientific research. In this study, we present MatterMind, a user-friendly agentic interface that closes the loop between first-principles computation, generative structure design, and large language model (LLM) analysis. The platform unifies (i) plane-wave first-principles workflows through Vienna Ab initio Simulation Package (VASP), (ii) crystal generation via diffusion models, and (iii) LLM assistance for error diagnosis, result interpretation, and report drafting. With simple “button-click” operations or natural-language prompts, users can: (1) build, launch, and monitor VASP jobs with automated parsing and recovery; (2) sample candidate crystals using modern generative models (MatterGen); and (3) obtain LLM-guided summaries, comparisons, and next-step suggestions for screening decisions. We illustrate end-to-end case studies that couple crystal generation to DFT relaxation and LLM-assisted assessment, reducing scripting overhead and improving transparency and reuse from a software engineering perspective. It provides an intelligent scientific tool with comprehensive software documentation, enhancing the efficiency and reliability of scientific exploration in chemical and materials science research.

1 | Introduction

In recent years, artificial intelligence (AI) and machine learning (ML) technologies are triggering a profound transformation in the research paradigm of materials science [1–3]. The data-driven approach to materials research has emerged rapidly. Its significance lies in its ability to efficiently explore the complex multidimensional materials design space, which is difficult to achieve with traditional methods based on experimental trial and error or pure theoretical simulation. Numerous examples have demonstrated the value of the data-driven [4–8] approach. In aspects, such as materials structure design, composition and

process optimization, autonomous experiments, and intelligent computing, machine learning can be used to reveal new laws and accelerate innovation. Compared with the previous means that rely on experience and cumbersome experiments, using machine learning models to conduct materials screening and property prediction in the virtual space has significantly improved efficiency and has promoted the discovery of many new function materials. Earlier than the current surge of AI-driven materials research, the United States launched the Materials Genome Initiative (MGI) in 2011, aiming to accelerate materials discovery and deployment through closer integration

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of experiments, theory, computation, and data infrastructure [9–11]. In this sense, MGI can be viewed as an important early milestone in data-centric materials research, whereas recent advances in AI and large language models have further expanded the scope and efficiency of this paradigm [12–14].

In the research of materials informatics [15–17], a rich set of machine learning algorithms and model libraries have been developed, which can be used to handle different types of materials problems. For example, algorithms, such as artificial neural networks, random forests, gradient boosting decision trees, naive Bayes, and support vector machines, have been successfully applied to construct models for predicting material properties. Once reliable experimental or computational data are obtained, these models can be used to quickly predict material performance in the virtual design space and assist in screening potential candidate materials for reverse design. However, the performance of the models highly depends on the selection of material descriptors. Descriptors are carriers that transform the composition, structure, and microscopic characteristics of materials into numerical features and are the basis for machine learning models to establish the structure–property relationship. To construct an accurate prediction model, it is necessary to provide a set of descriptors [18–21] that can effectively characterize material properties. For example, for inorganic crystal materials, common descriptors include elemental properties (such as atomic radius, electronegativity), unit cell symmetry, and structural topology. For organic molecules or organic-inorganic hybrid materials, molecular descriptors, such as molecular fingerprints and functional group characteristics, are required. In many advanced materials (such as metal-organic frameworks or hybrid perovskites), the organic and inorganic components jointly determine the material performance. After obtaining the descriptors, the further feature selection process is crucial. Selecting the most relevant subset from a large number of candidate descriptors through statistical or algorithmic methods can reduce the interference of redundant information on the model, improve the prediction accuracy, and enhance the interpretability of the model. These steps together ensure that the machine learning model not only has good generalization performance but also can reveal the internal relationship between material structure and performance.

Despite the great potential shown by data-driven materials design, the software tool ecosystem currently used in materials research still has obvious problems of fragmentation and heterogeneity [22, 23]. Software with different functions often belongs to their respective platforms and formats, lacking a unified and integrated working environment [24, 25]. This fragmented situation of multiple platforms and tools leads to data silos and collaboration obstacles: Researchers need to spend a great deal of energy consuming data and processes between different software, making it difficult to complete the overall research tasks in a timely and efficient manner. At the same time, the interfaces and operation methods of different software vary, increasing the operation complexity and learning cost, which is not conducive to researchers focusing on the problems of materials themselves. Especially in the field of materials informatics, the lack of user-friendly tool interfaces has seriously restricted the popularization of data-driven methods in daily scientific research. In contrast to the life science field, which has

thousands of easy-to-use web servers and visualization tools, the tool ecosystem in materials science is still in its infancy. There are very few web applications that can be used directly, and most of them are mainly data platforms. This lack has led to many existing materials algorithms and models not being widely adopted because they are difficult to deploy. Therefore, it has become an urgent need to build a more automated and integrated digital research platform for materials research.

Meanwhile, poor usability is also one of the prominent bottlenecks of current materials design software [26–28]. Many advanced materials calculation and machine learning programs need to run in a command-line interface or professional environments, such as Linux, which requires a high level of programming and system operation skills from users. This poses a high threshold for researchers who are proficient in experiments: they often need to spend months learning Linux system commands, scripting languages, and various programming libraries to master the basic software usage methods. This learning cost has greatly delayed the popularization and application of machine learning in the materials field and has also deterred a considerable number of researchers from data-driven tools. To address these issues, the research community has started to explore zero-programming or low-threshold materials calculation platforms [29]. For example, the MaterialsAtlas web [30] application [30] provides a series of materials discovery tools. Users can directly perform operations, such as material composition validity checks, property predictions, and searches for hypothetical materials on the web interface, thus reducing the usage threshold. Another example is the MLMD platform [31] which offers a “zero-programming” one-stop materials design interface. It supports a complete process including feature engineering, model training, optimization design, and result interpretation. Materials design tasks can be efficiently completed through a graphical interface. The NJmat software is an integrated tool that can achieve functions such as data preprocessing, feature generation, machine learning modeling, and visual analysis through simple button clicks in a Windows environment. It also significantly reduces the technical threshold for applying machine learning in the field of materials science. These efforts indicate that building user-friendly intelligent software for materials science is both feasible and necessary. It will help more researchers conveniently utilize data-driven methods to advance materials research. However, there is still a lack of LLM-assisted materials design software that natively integrates cutting-edge AI techniques with materials simulation tools. There is a need in the field of materials science for a new platform that integrates advanced algorithms and user-friendly interfaces to balance high-precision computational models and efficient and convenient user experiences for materials design.

In this study, a user-friendly LLM + Materials software architecture “MatterMind” is presented, which integrates materials calculation tools and LLMs, and constructs an intelligent closed-loop framework of “computation-generation-analysis” to achieve highly automated and intelligent support for the materials research and development process. The system adopts a loosely coupled and modular architecture and consists of three core functional modules: (i) A first-principles calculation engine based on *VASP*, which uses the *py4vasp* interface to optimize

parameters and control the task flow of materials quantum chemistry calculations; (ii) a cross-scale materials generation module driven by *MatterGen*, which combines the *pymatgen* library to parse and generate crystal structures; and (iii) an intelligent analysis module integrating multiple large language models. The intelligent analysis module integrates both cloud-based and locally deployed large language models. In the current platform, representative backends include Qwen-family models, Llama-4, and locally deployable reasoning models for offline or private analysis. Through the collaborative operation of the above modules, MatterMind achieves a connection between physical simulation and high-efficiency data analysis. It uses first-principles calculations to ensure the reliability of the results, utilizes the generation module to quickly explore the materials space, and the LLM module summarizes and interprets complex results in natural language to achieve a human-computer interactive analysis process. This design enables experimental researchers to conveniently perform material structure generation, simulation calculation and result analysis by simply setting a small number of parameters and clicking “execute” on a unified platform. The calculation process supports real-time monitoring, and the calculation results and analysis conclusions will be automatically presented to users in an intuitive graphical and text format. Compared with the traditional process that requires multiple software and manual data analysis, MatterMind reduces the operation difficulty and time cost and improves the efficiency and accessibility of materials research.

2 | Methods

2.1 | System Architecture and Front-End

MatterMind adopts a modular four-layer architecture for materials science workflows (Supporting Information S1: Figures S1–S4 and Figure 1a), consisting of the presentation layer, business logic layer, data access layer, and data storage layer. The presentation layer provides a web-based user interface for task submission, visualization, and interaction. Users can upload VASP input files (INCAR, POSCAR, POTCAR, and KPOINTS) to trigger automated first-principles calculations, or submit structure files (e.g., CIF format) for crystal generation via the MatterGen engine [32]. The business logic layer integrates the core functional modules of the platform, including VASP-based density functional theory (DFT) computation, MatterGen-based crystal structure generation, and LLM-assisted analysis. These modules communicate through RESTful APIs and real-time channels to ensure traceable workflows, responsive status updates, and extensibility. The data access layer manages data exchange between computation modules and storage systems, including data parsing, read/write operations, and synchronization. The data storage layer is responsible for persistent storage of input files, calculation outputs, and analysis results, implemented using MongoDB together with file-based storage across local and cloud environments. Upon task submission, the system automatically organizes computational outputs into structured formats, which are then analyzed by the AI module. The platform supports both cloud-based LLM APIs (e.g., Qwen-Max and Llama-4) and locally deployed models (e.g., DeepSeek-R1), enabling automated interpretation and structured reporting of computational results (Figure 1b).

2.2 | LLM, Metadata, and Data Management

To lower the LLM interpretation burden, MatterMind supports dual LLM backends: (a) local inference via Ollama (HTTP API) for on-prem offline analysis and (b) cloud inference via Alibaba Cloud Model Studio/DashScope for the Qwen family. The layer accepts structured outputs and logs from the compute layer and returns natural language summaries, error diagnoses, and follow-up suggestions; model selection (local vs. cloud) is configurable per task. Metadata (job descriptors, users, statuses, model choices, and file manifests) are persisted in MongoDB as document records, whereas heavy binary artifacts (input files, wavefunctions, vasprun/OUTCAR, generated CIFs, and AI reports) are stored on the server file system for robustness and efficient I/O. A document-oriented store is selected for its flexible JSON/BSON schema and natural mapping to hierarchical materials data; the approach aligns with common practice in materials informatics stacks (e.g., *pymatgen-db*). Data flow and execution are as follows: (i) The user uploads input files from the UI; the server validates, stores metadata in MongoDB, and stages files to localized storage. (ii) The scheduler invokes either VASP (DFT) or MatterGen (crystal generation). Upon completion, outputs are re-indexed and linked to the job record. (iii) The AI analysis layer consumes selected artifacts and returns structured summaries and human-readable reports. (iv) The frontend streams progress and surfaces results (including MatterGen's optional visual exports such as structure previews and simple distribution plots) via Socket.IO/WebSockets. To select practical LLM backends for integration into MatterMind, we performed a comparative evaluation of candidate models using a six-dimensional rubric covering factual accuracy, logical reasoning, creativity, domain adaptability, technical performance, and user experience (Supporting Information S1: Tables S1–S4). We note that this evaluation is intended to assess the quality of LLM-assisted analysis within the platform rather than to claim a rigorous physics benchmark. In the revised manuscript, we further supplemented this comparison with quantitative workflow-level evaluation, including average analysis scores from batch inference and time-cost comparison between the MatterMind automated pipeline and conventional multi-tool workflows. More detailed settings and results are provided in the Supporting Information. In both the VASP and MatterGen workflows, MatterMind does not directly pass raw unfiltered outputs to the LLM. Instead, key numerical and structural information is first extracted through tool-assisted parsing and consolidated into structured intermediate files (e.g., JSON-formatted metrics and metadata). The LLM is then prompted with task-specific system instructions to analyze only the provided computational facts, summarize the results, diagnose potential issues, and answer follow-up questions. In this design, the LLM functions as an interpretation and interaction layer rather than a replacement for first-principles calculations or trusted reference databases.

2.3 | VASP Computation and AI Analysis Module

The VASP module exposes a browser-based workflow that mirrors VASP's canonical input/output conventions while adding real-time observability and downstream AI-assisted interpretation. At submission, the user uploads the standard input files,

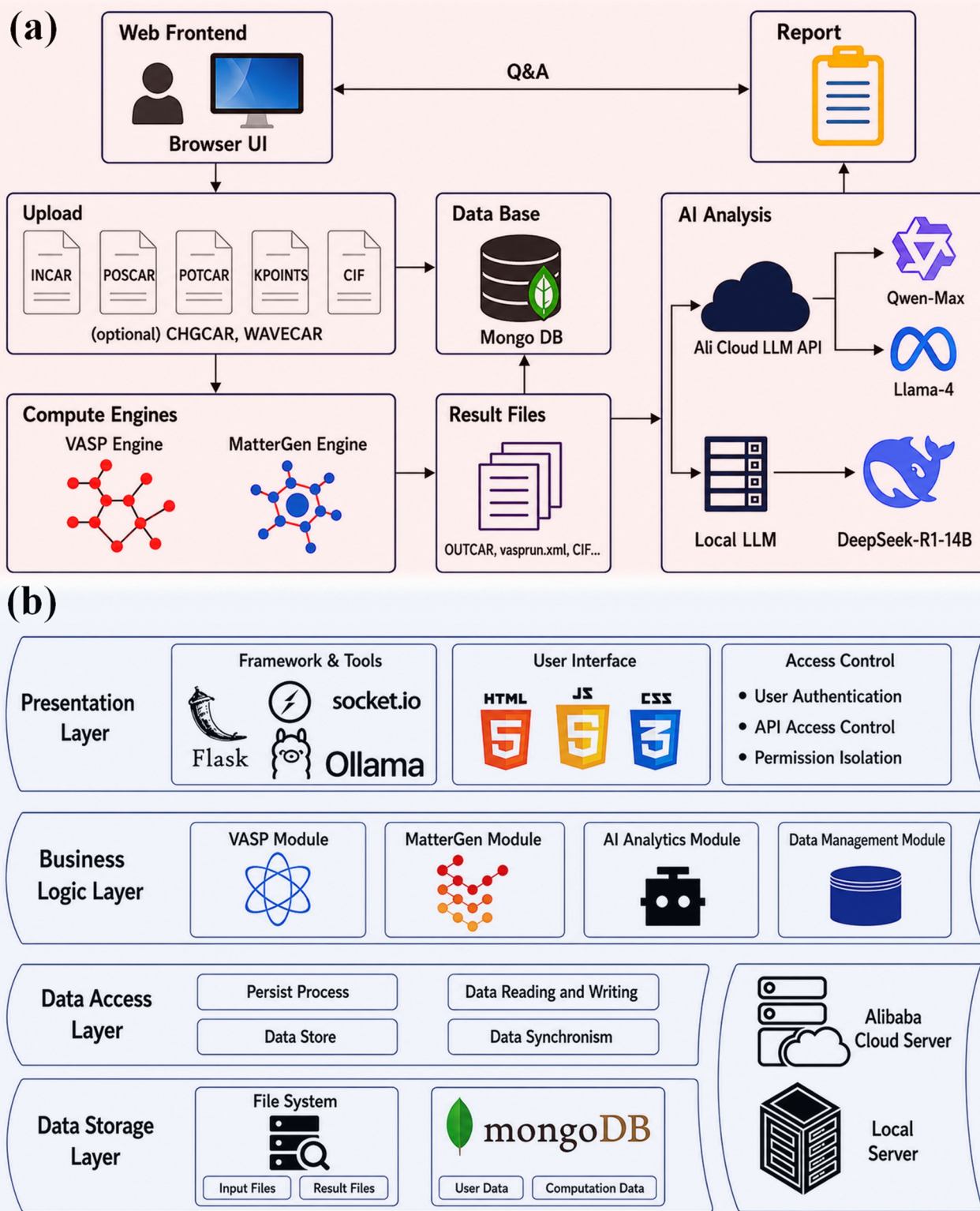


FIGURE 1 | Overview of the MatterMind platform. (a) Web-based workflow for VASP computation, MatterGen generation, and AI-assisted analysis. (b) Four-layer system architecture of MatterMind, including the presentation layer, business logic layer, data access layer, and 7 data storage layer.

such as INCAR (parameters), POSCAR (structure), POTCAR (pseudopotentials), and KPOINTS (k-point mesh), with optional files CHGCAR (charge density) and/or WAVECAR (wavefunctions) when a restart or post-SCF reuse is desired. The server

validates the presence and basic integrity of the minimal inputs and accepts the optional files for faster or more stable restarts (e.g., hybrid-functional or post-SCF workflows that rely on prior wavefunctions/charge density).

After acceptance, the backend provisions an isolated working directory, materializes user files, and launches the VASP job. During execution, stdout/stderr and iterative energies are tailed and persisted; core scientific results are generated in the standard VASP outputs (e.g., OUTCAR, OSZICAR, and vasprun.xml/vaspout.h5 when enabled). These outputs contain the full record of control settings, electronic-step convergence, Kohn–Sham eigenvalues, stress tensor, forces, local charges/moments, and other derived properties used in subsequent analysis. To provide experiment-style transparency, the frontend opens a WebSocket connection to the server streaming progress messages and log lines. This full-duplex channel is used to surface queueing/running states, electronic/ionic iteration summaries (e.g., from OSZICAR), and any warnings that require early user intervention. When the task completes, the results pane presents a file browser for selective download of the produced artifacts.

For interpretation, the user selects an LLM backend and triggers “AI Analysis.” The analyzer reads OUTCAR and the server-captured vasp.log (and, when available, machine-readable containers such as vasprun.xml/vaspout.h5) to summarize run conditions, sanity-check convergence, and extract salient physical quantities (e.g., band gap, total energies, forces, and magnetic moments) into a human-readable report. Where structured extraction is preferred, the pipeline can invoke py4vasp to programmatically access band structures, density of states (DOS), and energy trends from HDF5/XML without brittle text parsing; the generated summary is then used to seed a multiturn Q&A session for deeper problem-specific discussion. The system also supports interactive follow-up queries after file upload and calculation completion, allowing users to request explanation of INCAR settings, convergence behavior, and possible parameter adjustments in a conversational manner. In addition, the platform can distinguish common VASP-related workflows from uploaded file patterns and output signatures, enabling more context-aware downstream interpretation. Overall, this module keeps users within a single web interface from submission to insight: upload canonical VASP inputs (with optional CHGCAR/WAVECAR for restarts), monitor runs live via WebSockets, retrieve standard VASP outputs upon completion, and immediately transition to LLM-assisted reading and interactive follow-up, reducing turnaround time between calculation and scientifically meaningful interpretation.

2.4 | MatterGen Generation and AI Analysis Module

In the current web interface, users can switch among nine supported MatterGen checkpoints through a front-end selection menu, including both unconditional and property-conditioned variants, such as mattergen_base, mp_20_base, chemical_system, space_group, dft_mag_density, dft_band_gap, ml_bulk_modulus, dft_mag_density_hhi_score, and chemical_system_energy_above_hull. Because MatterMind is designed as a web-based platform, arbitrary front-end loading of user-supplied local models is not supported in the public online deployment; however, the open-source local version allows advanced users to reconfigure custom backends following the provided technical documentation. The user first selects one of the pretrained checkpoints released with MatterGen (including unconditional

and property-conditioned variants) and specifies the sampling hyperparameters batch_size and num_batches, which together determine how many candidate crystals are drawn in a single run. Optional conditioning is provided via a JSON dictionary passed as properties_to_condition_on (e.g., target magnetic density, band gap, bulk modulus, or energy above hull), together with a diffusion_guidance_factor that controls the strength of classifier-free guidance (larger values enforce closer adherence to the requested property at the expense of sample diversity). On submission, the backend calls mattergen-generate with the chosen checkpoint and parameters, yielding the standard MatterGen outputs: a generated_crystals.extxyz file in which each frame represents one generated structure, and a ZIP archive of CIF files that we unpack into a generated_crystals_cif directory for downstream use. The AI analysis pipeline then parses generated_crystals.extxyz alongside every .cif in generated_crystals_cif, extracting lattice parameters, composition, coordination environments, and simple stability descriptors into a structured human-readable summary, which in turn seeds multi-turn user-LLM interactions where the user can ask follow-up questions about prototype assignment, phase-diagram context, or potential functional applications of each generated crystal.

3 | Results and Discussion

3.1 | Case 1: VASP Calculation and AI-Assisted Interpretation

By simple front-end operations, VASP relaxations can be launched and monitored within MatterMind (Supporting Information S1: Figures S5–S9 and Tables S5–S8). To validate the end-to-end workflow, crystalline Si (diamond) and Au (fcc) were first employed as basic sanity-check examples to verify the correctness of the end-to-end VASP workflow (INCAR, POSCAR, POTCAR, and KPOINTS) submitted through the web interface. Upon geometry optimization, the Si–Si bond length increased slightly from 2.3572 to 2.3614 Å ($\Delta = +0.0042$ Å), whereas the Au–Au bond length decreased from 2.9496 to 2.8982 Å ($\Delta = -0.0514$ Å). These optimized values agree well with established reference data, confirming that the workflow yields physically reasonable structures under routine settings. Such modest geometric variations are typical for standard VASP optimizations, where atomic positions and, depending on the selected control parameters, lattice degrees of freedom are iteratively relaxed toward local equilibrium. To further demonstrate the capability of MatterMind beyond elemental benchmarks, a multicomponent Au–Si alloy (SiAu₃) was investigated using the same automated workflow. Compared to pure Si and Au, this system introduces greater compositional and local structural complexity, providing a more stringent test of automated job submission, output parsing, AI-assisted interpretation, and integrated workflow execution. As shown in Figure 2a,b and Supporting Information S1: Figure S10, the characteristic nearest-neighbor distance in SiAu₃ decreases from 3.2857 to 3.2416 Å after relaxation, indicating that the platform can reliably handle structurally complex systems within a unified framework. This example more closely reflects realistic application scenarios in materials research. For completeness, the electronic structures of the optimized systems were further analyzed. The total density of states (DOS) for Si, Au, and SiAu₃ are presented in Figure 2c. Si exhibits a clear depletion of states

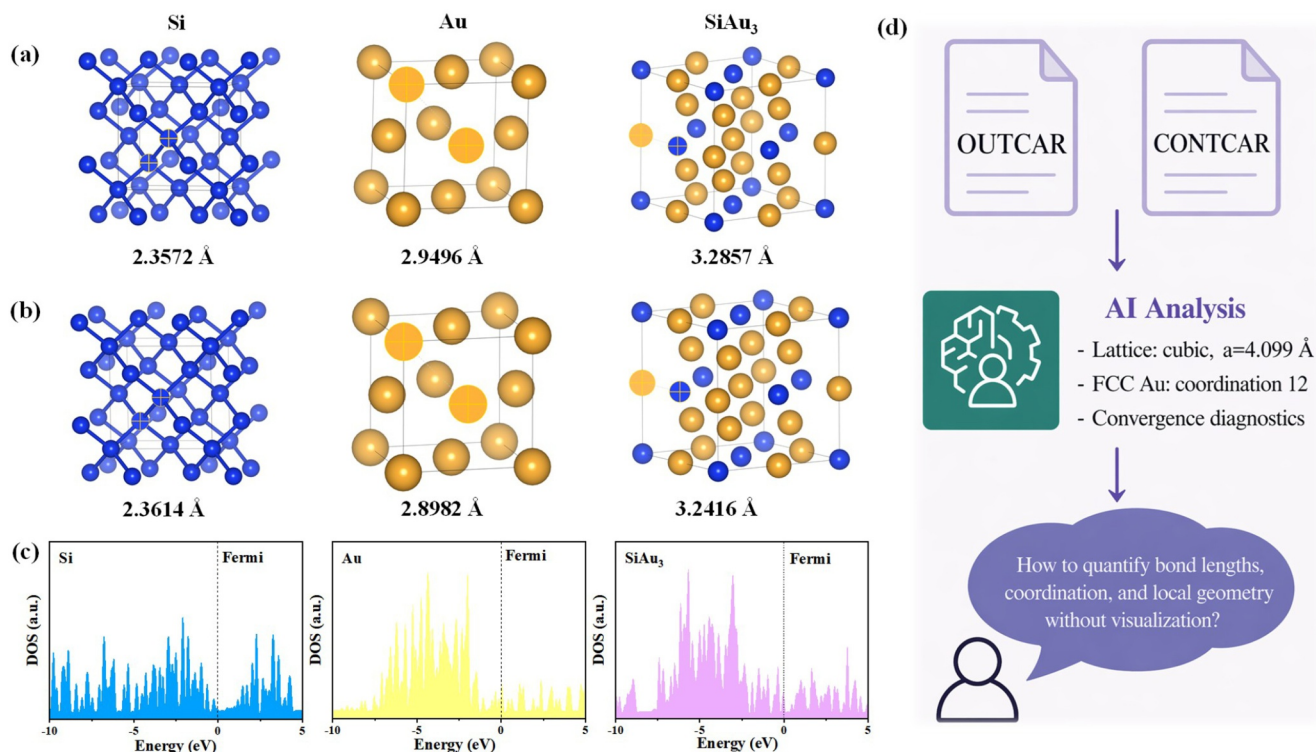


FIGURE 2 | (a) The bond lengths of Si, Au, and SiAu₃ before optimization using MatterMind. (b) The bond lengths of Si, Au, and SiAu₃ after optimization. (c) DOS of the optimized Si, Au, and SiAu₃. (d) Examples of automated AI-assisted interpretation after VASP computation.

near the Fermi level, with valence states predominantly located below 0 eV, consistent with its semiconducting nature. In contrast, Au shows a finite DOS at the Fermi level along with a broad distribution of occupied states at lower energies, characteristic of metallic behavior. The SiAu₃ alloy similarly displays a nonzero DOS near the Fermi level, indicating metallic-like electronic properties distinct from pure Si. Overall, the reproduced DOS features are consistent with the expected qualitative electronic characteristics of these systems, supporting the reliability of both the structural optimization and subsequent self-consistent calculations.

Beyond numerical outputs, MatterMind can automatically extract key structural and electronic descriptors from VASP results and organize them into structured reports for downstream AI-assisted analysis (Figure 2d). This functionality enables users not only to perform routine relaxation calculations via the web interface, but also to rapidly obtain interpretable summaries of lattice characteristics, local coordination environments, and convergence behavior for both simple and multicomponent systems.

Beyond the raw geometric changes, the AI analysis layer was used to interpret the corresponding OUTCAR for the Au relaxation. The model automatically identified the key calculation settings (PAW_PBE pseudopotential, 500 eV cutoff, k-point spacing of $\sim 0.04 \text{ \AA}^{-1}$ giving 20 k points, PBE-DFT, and 44 electrons) and summarized the convergence quality in terms of total free energy, residual forces, and negligible total drift, confirming that the optimization had reached a well-defined minimum. From the final lattice constant of $\sim 4.099 \text{ \AA}$ and the vanishing forces, the AI concluded that the relaxed configuration represents

a stable ground-state geometry suitable for further property calculations. Based on this, a follow-up query asked the AI model in MatterMind to infer the likely crystal prototype and to outline how bond lengths, coordination, and local structure could be analyzed directly from OUTCAR/CONTCAR without any visualization software as is shown in Figure 2d. The model correctly attributed the structure to bulk fcc Au, citing the near-ideal lattice constant, the characteristic fractional atomic positions (0, 0, 0), (0, $\frac{1}{2}$, $\frac{1}{2}$), ($\frac{1}{2}$, 0, $\frac{1}{2}$), ($\frac{1}{2}$, $\frac{1}{2}$, 0), and the cubic symmetry as evidence. It then sketched a rigorous text-based workflow for structural analysis: converting fractional to Cartesian coordinates using the final lattice vectors, computing nearest-neighbor distances (recovering the $2.898 \text{ \AA} \approx a/\sqrt{2}$ bond length), counting neighbors within a physically motivated cutoff to confirm the 12-fold coordination of fcc, and deriving bond-angle distributions and simple distortion metrics (e.g., variance of bond lengths or RMS deviation from an ideal fcc template). In combination with the Si and Au bond-length and bond-angle changes, this interaction illustrates how LLM can bridge from numerical VASP output to prototype identification and quantitative local-structure analysis in a way that is both physically consistent and immediately actionable for subsequent studies.

3.2 | Case 2: MatterGen Computation and AI-Assisted Interpretation

In a test case study, the MatterGen module (Supporting Information S1: Figure S11–S12 and Table S9–S12) in MatterMind perform geometrical optimization and yield nine structurally distinct crystalline candidates that span low-symmetry and more ordered motifs, which are rendered as representative structure snapshots in Figure 3a. For every sample, the

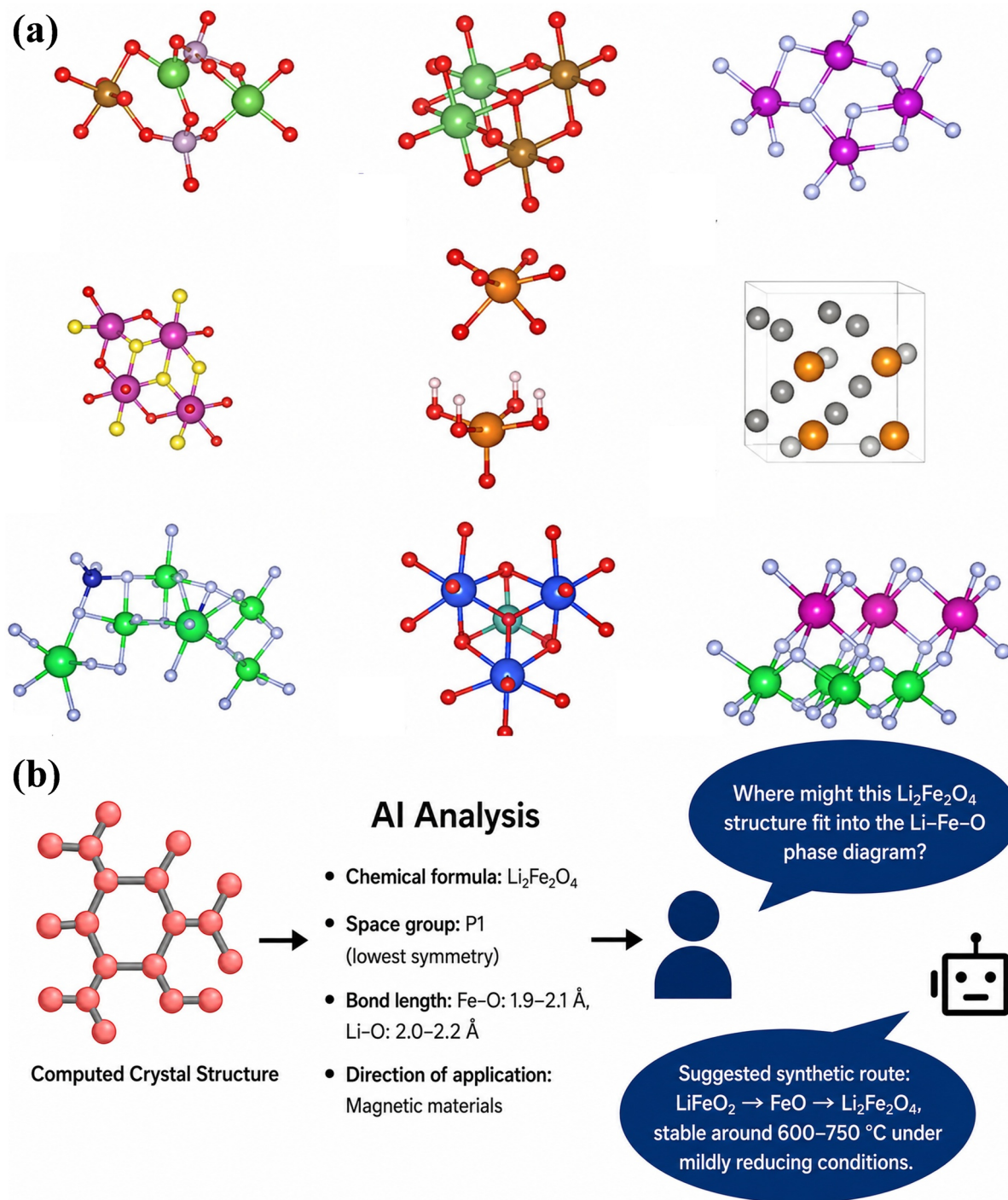


FIGURE 3 | (a) Examples of crystals generated by the MatterGen module. (b) An example of AI-assisted interpretation on MatterGen computation results.

workflow exports the raw MatterGen output from generated_crystals.extxyz to individual .cif files and register them in the database, providing a consistent starting point for the subsequent LLM-based structural and thermodynamic analysis described in the following section.

For a representative case study, one generated $\text{Li}_2\text{Fe}_2\text{O}_4$ candidate was selected from generated_crystals.extxyz and the corresponding gen_7.cif was passed to the LLM (DeepSeek-R1-14B) for post hoc analysis. Given only text-form structural data, the

model first verified that the lattice metrics extracted from the two files were identical and chemically consistent, confirming a triclinic P1 cell with FeO_6 and LiO_6 coordination polyhedra and Fe–O/Li–O bond lengths in the expected ranges for lithium iron oxides used as battery electrodes. It then placed this composition in the broader Li–Fe–O landscape by comparing the local environment and anisotropic cell parameters to known rocksalt and layered LiFeO_2 polymorphs, spinel-type LiFe_5O_8 , and layered Li_2FeO_3 , concluding that the structure is best viewed as a low-symmetry layer-derived variant rather than a conventional

spinel. As shown in Figure 3b, in a follow-up query on phase-diagram context and synthetic accessibility, the LLM model qualitatively located $\text{Li}_2\text{Fe}_2\text{O}_4$ in a mixed-valence region of the Li-Fe-O triangle neighboring LiFeO_2 - and Li_2FeO_3 -like compositions, and suggested that such a phase would most plausibly be stabilized as a metastable layered derivative under intermediate temperatures and moderately reducing oxygen chemical potentials, analogous to conditions commonly used for mixed-valence lithium iron oxides. The same interaction further highlighted potential functionality (cathode behavior enabled by $\text{Fe}^{2+}/\text{Fe}^{3+}$ redox, as well as possible magnetic or catalytic applications), mirroring the roles of related Li-Fe-O phases reported in the literature. Taken together, this example illustrates how the MatterGen-LLM pipeline can move beyond simple geometry checking to deliver chemically sensible prototype classification, phase-diagram reasoning, and synthesis hypotheses directly from machine-generated structures.

At the same time, the present study has clear limitations: the demonstrations are based on a small set of systems and rely on a single primary DFT code and one family of diffusion models; the LLM analyses, although chemically sensible, are not yet subject to systematic benchmarking for hallucinations, uncertainty, or robustness across broader materials scientists. Therefore, future work will expand the platform along three directions: (i) broader physics coverage, including phonons, finite-temperature dynamics, and alternative DFT engines; (ii) larger more diverse test suites to quantitatively evaluate both generative quality and LLM-based interpretation against established baselines for inorganic material discovery; and (iii) tighter integration of tool calling, information retrieval, citation, and uncertainty quantification so that more functionalities can be achieved and AI-generated explanations remain traceable to underlying data and literature.

4 | Conclusions

An integrated software platform is developed that couples DFT workflows and diffusion-based crystal generation with LLMs assistance for materials research. The current integration of LLMs with materials software demonstrates a well-structured approach to materials domain software engineering, serving as a component in supporting the Materials Genome engineering. Positioned against the emerging AI-for-materials landscape, MatterMind follows recent efforts that combine DFT engines, such as VASP, generative models, such as MatterGen, and LLM agents to accelerate simulation setup, structure generation, and knowledge extraction. The VASP module orchestrates standard solid-state calculations from canonical input files streams convergence information. A case study is demonstrated to expose optimized structures whose bond lengths, bond angles, and densities of states around the Fermi level closely match literature behavior, providing reliable ground-state geometries and electronic structures. The diffusion module wraps diffusion checkpoints for inorganic materials design, enabling unconditional and property-conditioned sampling of candidate crystals, whereas the downstream LLM layer parses the generated outputs to recover lattice metrics, coordination environments and prototype hints, where the model proposes a layer-derived configuration and plausible phase-diagram context based solely on structural text. To sum up, the present study offers a

comprehensive and user-friendly “LLM + materials” software platform to assist material design and analysis.

Author Contributions

Kaiyang Xu: data curation. **Yujun Lu:** software, visualization, investigation, data curation. **Lifeng Liu:** funding acquisition. **Lei Zhang:** funding acquisition, conceptualization.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting Information S1: mgea70075-sup-0001-suppl-data.docx.